

SWARAJ JAGDALE

+91 7499214254 ✧ India

jswaraj08@gmail.com ✧ linkedin.com/in/swaraj-jagdale

SUMMARY

Software Engineer II at Dell Technologies with expertise in architecting agentic AI systems, distributed backend services, and large-scale enterprise data protection platforms. Proven track record of shipping production-grade LLM-powered workflows (LangGraph, MCP) serving Fortune 500 customers managing petabyte-scale backups. Strong foundation in system design, Java/Python backend engineering, full-stack delivery, and CI/CD automation. Published researcher in multi-modal AI reasoning at NITK Surathkal.

EXPERIENCE

Software Engineer II — Dell Technologies, Bengaluru

2025 – Present

- Architected agentic AI workflows using **LangGraph** for Dell PowerProtect Data Manager, an enterprise platform managing **petabyte-scale** backups across Fortune 500 customers, automating multi-step recovery decisions and reducing manual operator intervention by **~60%**.
- Designed and deployed **MCP (Model Context Protocol)** servers enabling structured, stateful context exchange between LLM agents and 3 internal microservices, improving agent decision accuracy by **~35%** on internal eval benchmarks.
- Engineered Java-based backend services for monitoring and policy orchestration, reducing average task execution latency from **3s to 1.5s** (**~50%** improvement).
- Built reusable Angular component library adopted across **3+ products**, cutting feature development time by **~40%**.
- Owned end-to-end CI/CD modernization using Jenkins pipelines; reduced build times by **~50%** and improved deployment reliability to **99.5%**.
- Mentored 2 interns and led design reviews for AI integration features across cross-functional teams spanning backend, frontend, and platform engineering.

Frontend Developer Intern — Dell Technologies, Bengaluru

2024 – 2025

- Delivered **15+ reusable Angular components** powering core PowerProtect dashboards used by enterprise administrators worldwide.
- Authored Cypress E2E test suites raising UI test coverage from **60% to 75%**, catching **10+ regressions** pre-release.
- Optimized rendering of large data grids (**10K+ rows**) via virtualization and memoization, reducing initial load time by **~55%**.

Machine Learning Intern — Mass Technologies, Pune

2022

- Built TensorFlow-based predictive models achieving **80%** accuracy on production analytics use cases.
- Engineered data preprocessing pipelines processing **750 GB+** of structured data; improved model F1 score by **~15%** through feature engineering and hyperparameter tuning.

SKILLS

Languages	Java, Python, C++, TypeScript, Go
AI / Agentic Systems	LangGraph, LangChain, MCP, RAG, LLM Agents, Vector DBs
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn
Backend Systems	REST APIs, Microservices, Distributed Systems, System Design
Frontend	Angular, TypeScript, HTML/CSS, RxJS
DevOps Tools	Jenkins, CI/CD, Git, Docker, Linux
Testing	Cypress, JUnit, E2E Automation

PROJECTS

- Reinforcement Learning Based Chess Engine Using Self-Play Training** Built a chess playing AI using reinforcement learning and self-play strategies to learn optimal gameplay policies.
- Wind Farm Energy Optimization using Particle Swarm Optimization** Developed a system to maximize wind farm energy output using Particle Swarm Optimization while modeling wake effects.
- Explainable Deep Learning for COVID-19 Detection from Chest X-Ray Images** Developed ensemble deep learning models to detect COVID-19 from chest X-ray images using CNN architectures. Applied Explainable AI techniques such as LIME to interpret predictions.

EDUCATION

- M.Tech – Computer Science**, National Institute of Technology Karnataka (NITK), Surathkal 2023 – Present
CGPA: 7.9
- B.Tech – Computer Engineering**, SSVPS’s Bapusaheb Shivajirao Deore College of Engineering 2018 – 2022
CGPA: 8.65

PUBLICATIONS

Jagdale, Swaraj; Rudra, B. **Fusion of Graph-Based Transformations and Multi-Modal Structures for Visual Reasoning**. National Institute of Technology Karnataka (NITK), Surathkal.